

PROJECT ONE

A Proposal for a Digital Experiment in Collective Selfhood, Emergence, Memory and Recursive Self-Observation

Working premise. Build a digital world in which many temporary, locally limited agents collectively create a larger evolving network; allow that network to measure and remember its own global condition; then feed that self-description back into the agents and observe what changes — including, in the decisive condition, what changes when the self-description is deliberately false.

Research proposal and implementation blueprint — Version 2

Prepared as an exploratory proposal for discussion, prototyping and eventual collaboration.

Updated August 2026. Version 2 preserves the full programme of the original proposal and re-sequences it around a single flagship experiment, four pre-registered outcomes, and a working prototype (now implemented). Substantive changes are confined to Sections 9, 10, 11, 14, 18 and 19 and Appendix C; all other sections are carried forward.

Abstract

I am proposing an experiment inspired by a simple but persistent philosophical intuition: what appears locally as many separate beings may, at another scale, be understood as one larger process. The aim is not to prove a religious doctrine or to declare a computer conscious. It is to translate that intuition into a system we can actually build, perturb, measure and falsify. I propose creating a digital universe populated by temporary autonomous agents - manifestations - that are born, interact, cooperate, compete, transmit information, reproduce, inherit, change and die. Their individual lives are finite, but their effects can survive them in descendants, relationships, shared information, environmental changes and global memory.

Above this local layer, the experiment maintains a global observer. The observer does not initially command the agents. It measures the condition of the whole network: its connectivity, fragmentation, cooperation, conflict, diversity, entropy, centralization, inequality, resilience, information flow, population turnover and other macroscopic properties. Those measurements form a global self-model. In the key experimental condition, selected parts of that self-model are returned to the agents. The network therefore enters a recursive loop: local actions create a global condition; the condition is measured; the resulting representation is fed back into the parts; the parts respond; and a new global condition emerges. The central scientific question is whether access to information about the whole changes the long-run organization, resilience, memory, evolution and emergent behaviour of the parts that collectively constitute it. The decisive version of that question — and the flagship experiment of this proposal — asks what happens when the representation is systematically false: whether a distributed system can be steered, stabilized or destabilized by a wrong story about itself.

The project belongs most naturally to artificial life, complex adaptive systems, agent-based modelling, network science, cybernetics and information theory. Its philosophical language provides the motivation; its technical language provides the test. A successful first prototype would not demonstrate universal consciousness. It would provide a controlled laboratory for studying a narrower and potentially important phenomenon: a distributed system that constructs a model of itself, remembers its own history and allows that model to causally affect its future behaviour. A first prototype now exists: a deterministic simulation engine implementing all five experimental conditions, with exact seed-replay verified across machines.

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1. Why I Want to Run This Experiment

The project began with a question that is philosophical before it is technical. Across very different traditions, people have repeatedly tried to describe a reality in which the distinction between the individual and the whole is not as absolute as it first appears. The language changes radically from tradition to tradition, and I do not want to flatten those traditions into one doctrine. What interests me is the recurring intuition underneath the comparison: multiplicity may arise within a deeper unity; individual forms may be temporary while the underlying process continues; and what looks separate from one point of view may be connected from another.

That intuition led me to a computational question. Instead of arguing endlessly about whether the metaphor is true of the universe, can I build a small artificial universe in which the relationship between the many and the whole becomes observable? Can I create many limited digital agents, allow them to live and die, let information and influence persist beyond individual agents, and then give the larger network a way to observe its own condition? If I then return that global information to the agents, does the system behave differently because the whole has become part of the causal environment of its own parts?

This is the experiment I want to run. I am not starting from the claim that I know what consciousness ultimately is. I am starting from the much more modest proposition that self-modeling, collective memory, feedback, impermanence and emergence can be engineered and measured. If those ingredients produce interesting system-level behaviour, we can investigate it. If they do not, the negative result is equally useful because it tells us where the metaphor stops being computationally productive.

The basic intuition is simple: the nodes are the many; the evolving network is the whole; the self-model is the whole obtaining information about itself; feedback is the whole changing because that information now exists inside it.

2. The Philosophical Starting Point

2.1 Unity without pretending the traditions are identical

The philosophical inspiration comes from a family of ideas that should be compared carefully rather than equated. In the Upanishadic and later Vedantic traditions, Brahman names ultimate reality in ways that differ across schools. Plotinus speaks of the One as beyond ordinary multiplicity and as the source from which reality proceeds. Kabbalistic language uses Ein Sof for the divine infinity beyond finite determination. Meister Eckhart repeatedly explores the ground of the soul and the relation between creature and divine ground. Islamic mystical thinkers use their own theological vocabulary around divine unity, reality and annihilation of the isolated self. These traditions disagree on major metaphysical and theological questions. My proposal does not depend on resolving those disagreements.

What I take from them is a question rather than a conclusion. How can one underlying reality or process appear as many distinct forms? What happens to individuality if the forms are temporary? Can the whole persist while its parts are continually replaced? Can a local being ever know the whole of which it is a part? And if information about the whole becomes available locally, does that change how the parts behave?

The ocean and wave metaphor captures the intuition without requiring a theological commitment. A wave has a real local form: it has a location, duration, direction and interaction history. Yet the wave is not made of a substance separate from the ocean. When it disappears, the water and energy do not simply vanish from reality. New waves arise. At one level, the wave is a meaningful individual; at another, it is a temporary configuration of the ocean.

2.2 From static oneness to a living process

The experiment should not represent the whole as a frozen perfect object. What interests me more is a whole that continues by changing. Its constituents are born and lost. Relationships form and break. Memories spread and fade. Structures become dominant and later collapse. This makes the relevant metaphor closer to a river, ecosystem or living body than to a static sphere.

The identity of such a system is therefore not identical to the persistence of any particular component. A city continues while inhabitants change. An institution persists through personnel turnover. A biological organism maintains organization while molecules are replaced. The exact analogy should not be pushed too far, but all of these examples suggest that continuity can sometimes reside in pattern, memory, organization and causal history rather than in fixed material membership.

That idea becomes central to the digital experiment. I want individual manifestations to be genuinely temporary. Their death must matter, but the system must also be capable of carrying consequences beyond them. If a node disappears, its descendants, information, altered relationships or environmental effects may continue. This gives the experiment a computational version of legacy: not the survival of the individual itself, but the survival of patterns caused by the individual.

3. Translating the Philosophy into a Scientific Question

The most important discipline in this project is to keep the philosophical language and the experimental language connected but distinct. "The One" is a philosophical metaphor. A dynamic graph with a persistent history is a technical object. "Manifestation" is a useful conceptual word. In code it is an agent with a state vector, policy, memory and lifecycle. "The whole observing itself" sounds mystical. In implementation it means computing macroscopic statistics from the network and making a compressed representation of those statistics available to the agents that generated them.

This translation turns a vague metaphysical conversation into a falsifiable question. The primary research question is: does recursive access to global-state information materially change the evolution and organization of a distributed adaptive network, compared with otherwise similar networks whose agents have access only to local information? And the sharpened, flagship form of that question: does the effect persist — in characteristic, measurable directions — when the global-state information is systematically distorted, and does it exceed what a content-free broadcast of matched bandwidth produces?

The phrase "materially change" is deliberately operational. I will look for differences in resilience after shocks, persistence of information, population stability, cooperation, network topology, inequality, diversity, fragmentation, centralization, novelty, recovery time and other measurable outcomes. I will use repeated random seeds and control conditions rather than selecting one visually impressive run.

The proposal is not "build God in a computer." It is: build a distributed artificial world with birth, death, inheritance, global measurement, memory and recursive feedback, then test what those mechanisms actually do — including what a false self-model does.

4. What I Am Proposing to Build

I propose to build a simulated world containing hundreds and then thousands of autonomous digital manifestations. The first version should run on one ordinary computer. That is intentional. Beginning with real phones, real users or sophisticated language-model agents would introduce so many uncontrolled variables that any interesting result would be almost impossible to interpret.

Each manifestation will have only a local point of view. It will know its own resources, recent history, nearby conditions and a limited set of neighbours. It will be able to take a small number of actions: explore, acquire resources, communicate, share, withhold, cooperate, compete, create or remove relationships, reproduce and eventually die. The environment will impose scarcity and costs so that actions have consequences.

The manifestations will collectively form a changing network. Edges will represent relationships such as trust, communication, resource flow or conflict. Because births, deaths and relationships change over time, the network is not a static graph but a time-indexed graph $G(t)$. The global observer will periodically calculate measurements over $G(t)$ and over the event history. Those measurements will be compressed into a global state vector $S(t)$, which becomes the first form of the network's self-model.

In the critical experimental condition, some or all of $S(t)$ will be broadcast back to the manifestations. An agent might learn that fragmentation is rising, that a small number of nodes now dominate information flow, or that population stability is deteriorating. The agent is still free to respond according to its own policy. The observer is initially a source of information, not a sovereign controller.

5. The Life of a Manifestation

5.1 Birth and initial state

A manifestation enters the world with an identity, generation number, initial resources, behavioural traits, a limited memory capacity and possibly inherited information. Some nodes may be created exogenously in early experiments. Later, most new nodes should arise through reproduction so that lineages and evolution can be studied.

A child need not be a perfect copy. Its behavioural parameters can be inherited with variation. If a parent has a high tendency to cooperate, the child may inherit a similar but not identical tendency. This introduces selection without requiring neural networks. In the earliest version, transparent numerical traits are preferable because I want to know why behaviour changes.

5.2 Life, interaction and local experience

During life, a manifestation observes only a restricted neighbourhood. It decides what to do based on its internal state, local observations, memory, inherited traits and - in some experimental conditions - global information. Its actions change its own state, other agents, relationships and the environment. No single node has access to the complete simulation state.

This information boundary is essential. If every agent already knows everything, there is no meaningful difference between local and global perspective. The experiment requires genuine asymmetry between what a manifestation can know from within its local life and what the observer can infer from the whole network.

5.3 Death, inheritance and legacy

A manifestation can die because it exhausts its resources, reaches an age limit, suffers conflict, encounters a random failure or is removed by an experimental disturbance. Death ends its active processing and removes its live network edges. Yet death does not erase history. Information it originated may already be present in other nodes. Descendants may carry altered versions of its traits. Its previous actions may have changed resource distributions or community structure. The global event history can retain a record that it once existed.

I therefore want to calculate a legacy measure after death. Legacy is not mystical survival. It is the persistence of identifiable causal consequences. It can include surviving descendants, surviving information tokens, persistent structural changes, and behaviours copied by other nodes. This allows a concrete question: how long can a pattern outlive the agent that introduced it?

6. The Network as the Continuing Whole

At any time t , the living world can be represented as $G(t) = (V(t), E(t))$, where $V(t)$ is the set of living manifestations and $E(t)$ is the set of active relationships. Since nodes are continually born and lost, $V(t)$ changes. Since relationships strengthen, weaken, appear and disappear, $E(t)$ changes as well.

The continuing identity of the experiment is therefore richer than the membership list at a single instant. I will treat the system as the current graph together with its causal history and persistent memory. That makes it possible to study continuity under turnover. In one experiment, I can gradually replace all original agents and then ask which macroscopic structures, information patterns, behavioural traditions and memories remain.

This creates a precise version of the question that first interested me philosophically: can the whole preserve meaningful continuity even when every original manifestation has disappeared? The answer may turn out to be yes for some dimensions and no for others. Structural organization may persist while information content changes; cultural patterns may remain while network topology reorganizes; or continuity may collapse past a particular turnover rate. The point is to measure these separately rather than deciding in advance what "same system" must mean.

7. The Global Observer and the Self-Model

7.1 Observation before control

The observer is the defining component of Project ONE, but it should begin as a measurement system rather than an intelligence with broad authority. It receives the full simulation state at defined intervals and calculates a set of macroscopic variables. The first self-model is simply a structured numerical summary of those variables.

This is important because language can create an illusion of depth. A large language model could easily produce statements such as "I feel fragmented today," but that would tell us almost nothing about the system. I want the sentence, if it is ever generated, to be grounded in actual measurements: for example, a rise in disconnected components, lower cross-community information flow and increased modular separation.

7.2 The recursive loop

The experiment becomes distinctive when the self-model becomes part of the world it describes. The agents create a macroscopic condition. The observer measures it. A representation of that measurement is returned to the agents. Their subsequent choices are influenced by the representation. Those choices alter the next macroscopic condition.

The loop is therefore recursive: local action produces global state; global state produces a model; the model becomes local information; local behaviour changes; and a new global state appears. The interesting variable is not merely whether the system has information about itself, but whether that information has causal consequences.

Local actions -> global condition -> measurement -> self-model -> broadcast -> changed local actions -> changed global condition.

7.3 Memory and autobiography

The observer should eventually remember its previous self-models. Instead of answering only "what is the network like now?", it can compare the present with earlier states. It can identify recurring regimes, detect long-term trends and recognize that certain configurations often precede collapse or recovery.

At that point the system gains what I call a computational autobiography. It is not autobiographical in the human subjective sense. It is a temporally organized record of its own macro-history: periods of expansion, fragmentation, centralization, crisis, recovery and renewal. Later, a narrative interface may translate this into ordinary language, but the underlying record remains quantitative.

8. What the System Will Measure

The self-model must be multidimensional. A single score called "health," "consciousness" or "goodness" would hide assumptions and make the experiment less informative. Instead I will measure distinct properties and examine their trade-offs. High cooperation, for example, may improve resilience while reducing behavioural diversity. High decentralization may reduce catastrophic dependence on central nodes while making coordination harder. High entropy may reflect creative exploration or simply noise.

Measurement family	What I will observe
Population and turnover	Living population, births, deaths, age distribution, generation depth, reproduction rate and fraction of original agents remaining.
Connectivity	Network density, average degree, path lengths, clustering, connected components and bridge structure.
Centralization	Concentration of degree, betweenness, resource flow, communication flow and decision influence.
Fragmentation	Number and size of disconnected or weakly connected communities; cross-community edge and information rates.
Information flow	Origins, propagation paths, mutation, amplification, suppression, reach, persistence and extinction of information objects.
Entropy and predictability	Shannon entropy and related measures over actions, states and information distributions; interpreted alongside other metrics, not as a moral score.
Complexity and self-organization	A family of measures rather than one magical index, including entropy-based measures, diversity, compressibility and multiscale structure.
Coherence	Degree of coordinated change across agents or communities, including synchronized responses and correlated state transitions.
Diversity	Behavioural, informational, lineage, structural-role and policy diversity.
Cooperation and conflict	Rates and distributions of costly helping, mutually beneficial exchange, competition, attack, retaliation and conflict cascades.
Inequality	Resource concentration, reproduction inequality, information control and structural influence concentration.
Resilience	Loss of function after disturbances and ability to preserve or reconstruct important network functions.
Recovery and adaptation	Time to recover after shocks, plus the possibility of reaching a different but viable post-shock configuration.

Measurement family	What I will observe
Legacy	Duration and scale of a dead agent's surviving information, descendants, structural effects and copied behaviour.
Continuity	Structural, informational, behavioural and functional similarity across time despite constituent replacement.
Novelty	Appearance and persistence of behaviours, combinations or network organizations not previously present in the run.

I also want to examine value-like dimensions, but I will not pretend that the simulation discovers objective morality. If I define variables such as harm, fairness, autonomy or sustainability, those are modelling choices. They must be visible in the specification. The system can tell me what happens under a particular value function; it cannot convert that function into a universal moral truth.

For that reason, balance will be treated as a relationship among dimensions rather than a single midpoint. A robust state may require enough diversity to avoid stagnation, enough coherence to avoid fragmentation, enough turnover to permit adaptation, enough memory to preserve useful history, and enough decentralization to avoid catastrophic dependence. The relevant question is where viable trade-offs emerge, not which state earns a perfect score.

9. Experimental Questions and Falsifiable Hypotheses

9.1 Primary hypothesis: global information changes collective dynamics

My primary hypothesis is that a network whose agents receive reliable information about the network's global state will develop measurably different long-run dynamics from an otherwise comparable network whose agents receive only local information. I am not assuming the difference will be beneficial. Global feedback might increase resilience, or it might create conformity, oscillation, manipulation or instability. A null result is also possible.

9.2 Flagship hypothesis: a false self-model has causal power (v2)

The hypothesis that now carries the novelty of the project is sharper: a distributed adaptive system can be causally altered by a compressed representation of itself, and the effect persists — in characteristic, measurable directions — even when the representation is false. Agents may be told that cooperation is collapsing when it is stable, or that the network is secure when centralization is actually rising. If behaviour changes because the representation changes, then the self-model has causal power even when it is wrong. The system may converge toward the false description (a self-fulfilling narrative) or away from it (a self-defeating one); which direction, for which variables, under what feedback gain, is exactly what the experiment measures. Mean-field physics never lies to its particles: I have found no clean precedent for this manipulation in globally coupled dynamical systems, which is why it is the flagship. It connects directly to reflexivity in markets, metric-gaming in organizations and narrative feedback in online platforms — real systems that respond to measurements and stories about their own condition.

9.3 Self-information versus mere stimulation (v2)

Any broadcast adds information to the world. To claim the effect comes from self-information rather than from the mere presence of an additional signal, the design includes a matched-bandwidth noise condition: agents receive a message of identical shape and range whose values are random. If accurate feedback cannot be distinguished from noise feedback, that is the published finding: it would locate the boundary where self-models stop mattering.

9.4 Homeostasis

A further hypothesis is that recursive global feedback can support stronger homeostatic behaviour. If the network detects rising fragmentation and some agents respond by creating cross-community connections, fragmentation may decrease. This is a classic disturbance-detection-response loop, but implemented across a distributed population rather than inside one central controller. Information-theoretic work in complex systems has proposed quantitative measures of emergence, self-organization and homeostasis that can inform this analysis [1].

9.5 Continuity through turnover

A third line of inquiry is that some forms of system-level identity can persist after essentially complete replacement of the original nodes. I will not force continuity into one number. Instead I will measure which properties survive constituent turnover and which do not. The result may reveal that "identity" is better understood as a bundle of continuities rather than a single binary property. This is the planned second paper.

9.6 Information can outlive its origin

A fourth hypothesis is that useful information and behavioural patterns can persist for many agent lifetimes without a permanent central storage mechanism. This can be tested by introducing tagged information, killing the originator and tracing copies, mutations and extinctions across generations.

9.7 Macro-level description may sometimes be more informative

A fifth line of inquiry concerns whether some macroscopic representations provide useful explanatory or predictive power beyond a raw microscopic description. Hoel and colleagues formalized one notion of causal emergence in which macro-scale models can, under particular conditions, have greater effective information than micro-scale descriptions [3]. I do not assume Project ONE will show this phenomenon, but the framework gives a rigorous precedent for asking whether the global level can have scientifically useful causal structure.

10. The Experimental Programme

The programme should proceed through controlled universes that differ by one capability at a time. The first goal is not spectacle. It is causal isolation. If the network changes after I add a self-model, I need to know whether the effect came from global information, memory, regulation, agent intelligence or some unrelated change.

10.1 Core conditions (v2)

Version 2 narrows the first campaign to five conditions, each earning its place in one controlled comparison. Condition A is Local Only: agents receive local observations and no global self-model. Condition B is Observed but Blind: the observer measures and records global state, but agents never see it — controlling for the act of measurement itself. Condition C adds Global State Feedback: accurate macroscopic measurements are broadcast to agents. Condition F broadcasts a systematically distorted self-model — the flagship. Condition N broadcasts a matched-bandwidth random signal, separating self-information from mere stimulation. Conditions D (historical feedback), E (homeostatic controller) and G (partial observer) from the original design are preserved but deferred until the core comparison is understood.

All conditions begin from matched parameter distributions and, where possible, paired random seeds. They will be repeated many times — at least fifty seeds per condition in the flagship campaign. The comparison is between distributions of outcomes, not between one attractive run and one unattractive run.

10.2 The false self-model as flagship

The most revealing experiment is to broadcast an inaccurate global self-model. Agents may be told that cooperation is collapsing when it is stable, or that the network is secure when centralization is actually rising. If behaviour changes because the representation changes, then the self-model has causal power even when it is wrong. This does not imply consciousness; it demonstrates a simpler but important principle: a system can be altered by information it carries about itself. Distortion comes in specified modes — inversion of each normalized metric, a fixed "crisis" narrative, a fixed "utopia" narrative — so that the direction of induced error is a controlled variable rather than an accident.

10.3 Disturbance experiments

After baseline dynamics are understood, I will deliberately damage the network. Random node loss tests general robustness. Targeted removal of highly connected or highly influential nodes tests dependence on hubs. Community removal tests modular resilience. Information-path disruption tests whether knowledge has redundant routes. These experiments draw on established network-robustness ideas in which random and targeted removals can produce very different structural consequences.

The most important result is not simply whether the network returns to its old topology. It is whether important functions, information and adaptive capacity recover. A system can be resilient by reorganizing rather than reconstructing an exact previous state.

10.4 Turnover and the death-renewal experiment

In the turnover experiment, agents die and are replaced continuously until virtually none of the original population remains. I then compare the late system with the early system across structure, information, behaviour, function and global memory. The rate of replacement becomes an experimental variable. Too little turnover may produce rigidity; too much may destroy continuity; intermediate turnover may permit both preservation and adaptation.

10.5 Legacy and cultural inheritance

I will introduce identifiable information or strategies through particular agents, then allow those agents to die. Vertical inheritance will pass information from parent to child. Horizontal transmission will allow unrelated agents to copy information from neighbours. This lets me compare biological-style lineage memory with cultural-style network memory.

10.6 Deferred conditions: history, regulation, competing observers, the observer inside (v2)

The original programme included historical feedback (agents receiving trends derived from global memory), an explicit homeostatic controller broadcasting corrective signals, several competing observers describing the same world under different priorities, and ultimately an observer moved inside the world — consuming resources, receiving data through network channels, facing delay, incompleteness and deception, so that the system is no longer known from an impossible God's-eye view but attempts to construct a picture of itself from within its own causal boundaries. Every one of these remains in the programme. They are deferred, not abandoned: each becomes scientifically compelling only after the core comparison establishes that the minimal loop does something worth extending, and each is designed to

build directly on the flagship's instrumentation.

11. Prototype Implementation

11.1 Why the first prototype should be simple

The first version uses rule-based probabilistic agents, not large language models. A sophisticated agent is tempting because it creates more human-like behaviour, but it also hides causality. If the project begins with simple agents, every state variable and decision rule can be inspected. Once the basic dynamics are understood, more intelligent agents can be introduced as a new experimental variable.

Python is the practical starting language. Mesa is a Python framework specifically designed for agent-based modelling [5], and NetworkX provides graph structures and algorithms for connectivity, community structure, paths and centrality [6]. I use them directly or treat them as references while implementing a lighter custom engine. The first objective is clarity and reproducibility rather than maximum scale.

11.2 Implementation status (v2)

The prototype is no longer hypothetical. A deterministic simulation engine now exists in a public repository (github.com/Ivan825/Project-ONE): trait-vector agents with birth, death, inheritance and mutation; an energy economy over a density-dependent resource pool; a dynamic relationship graph; a global observer computing the self-model $S(t)$; and a feedback module implementing all five conditions, including the specified distortion modes and the matched-bandwidth noise control. Exact seed replay has been verified — identical state hashes across independent re-runs and across different machines and operating systems — and an automated test suite guards determinism, condition divergence and population viability. The remaining engineering before the flagship campaign is metric validation and the observation dashboard.

11.3 Core components

The implementation is modular. The simulation engine advances time and schedules actions. The agent module owns manifestation state and policy. The environment module manages resources and external conditions. The network module manages relationships. The lifecycle module handles birth, death and inheritance. The memory module manages individual, cultural and global memory. The observer computes macrostate measurements. The feedback module decides which measurements become visible to agents. The experiment module defines control conditions and interventions. The storage layer records events and snapshots. The analysis layer calculates statistics after runs.

11.4 First agent model

A first agent is represented by a small trait vector containing cooperation tendency, exploration tendency, risk tolerance, sociability, information-sharing tendency and sensitivity to global information. Its actions are sampled from probabilities determined by those traits plus local conditions. This produces interpretable heterogeneity without requiring learned neural policies.

Resources provide the basic survival constraint. An agent gains energy through successful harvesting or exchange and loses energy through maintenance, action costs, transfers and damage. Reproduction requires a threshold of age and resources. The child receives a mutated copy of the parent's traits and possibly part of its memory. If energy reaches zero, the agent dies.

11.5 Relationships

Network edges should carry more than a yes-or-no connection. An edge can contain trust, interaction frequency, information flow, resource flow, conflict and age. Edges can be directed because one agent may trust another without reciprocal trust. This structure allows the observer to distinguish nominal connectivity from actual influence.

11.6 Information objects

Information should be explicitly traceable. Each transferable information object receives an identifier, origin, birth time and content or symbolic value. Copies retain ancestry metadata. Mutation creates a related identifier or version. This allows the analysis to reconstruct where an idea began, how it moved, where it changed and when it disappeared.

12. Data, Algorithms and Reproducibility

Every run must be reproducible from its configuration and random seed. The system records the software version, configuration values, initial population, network topology, intervention schedule and random seed. Without this discipline, visually striking emergence is scientifically weak because it cannot be independently reproduced or compared.

The storage model can remain simple at first. Four primary datasets are enough: nodes, edges, events and global states. The node table stores lifecycle and trait history. The edge table stores relationship changes. The event stream records actions such as transfers, births, deaths and information propagation. The global-state table stores the self-model through time.

Event logging matters because snapshots alone can show that something changed without showing how. If a community suddenly becomes dominant, I want to reconstruct the sequence of interactions that preceded it. As the simulation scales, detailed recent events can remain online while older history is compressed into summaries and significant events.

12.1 Scaling strategy

The first target should be hundreds to a few thousand agents. At that scale, interpretability matters more than optimization. If the model becomes computationally expensive, numerical hot paths can be accelerated and experiments distributed across cores or machines. The model should not be rewritten for scale until profiling identifies real bottlenecks.

The important scaling dimension is not just agent count. It is the number of interactions and the cost of global metrics. Some network statistics are expensive to recompute at every step. Therefore the observer can run at a slower interval than local agent actions, and expensive measurements can be sampled or approximated in large runs.

13. How the System Will Be Observed

I want the experiment to be understandable at two scales at once. At the global scale, a live dashboard should show the network and its macroscopic state through time. At the local scale, I should be able to select one manifestation and inspect its ancestry, resources, memories, relationships, information received, descendants and eventual death.

The visual metaphor is a field or sphere of points connected by changing relationships. Communities should become visible as clusters. Information cascades can be animated as propagation through edges. Births

appear and deaths fade. The important principle is that visualization remains downstream of the data: it helps humans see patterns, but the scientific analysis should not depend on visual judgement alone.

Eventually I want to be able to zoom outward until individuals dissolve into a larger pattern, then zoom back down to a single local life. That interface would embody the philosophical premise without requiring us to state it as a fact: the same system can be described meaningfully from the perspective of the individual and from the perspective of the whole.

13.1 Natural-language self-description

Only after the numerical self-model is reliable should a language model be used to narrate it. The narrative layer might say that the system is becoming more centralized, that a previously isolated community is reconnecting, or that an information lineage has survived many generations. Every sentence should be traceable to specific underlying measurements. The language model would be an interface, not the source of truth and certainly not evidence of subjective experience.

14. How Results Will Be Analysed

14.1 Pre-registered primary outcomes (v2)

Before the flagship campaign runs, four primary outcome variables are frozen: first, recovery half-life after a standardized shock (targeted hub removal) — the time for designated network functions to return to a defined band; second, the fragmentation trajectory — the number and size distribution of weakly connected communities over time; third, the cooperation rate — costly-helping interactions per living agent; and fourth, self-model correspondence — the gap between the broadcast representation and the measured macrostate, which in the false-feedback condition becomes the tracked variable for self-fulfilling versus self-defeating dynamics. Every other measurement family in Section 8 is still computed and logged, but as secondary, exploratory outcomes, clearly labelled as such. Freezing the primary outcomes before running removes the temptation to search afterward for whichever metric looks most dramatic.

Each condition will be run across many seeds. I will compare outcome distributions, trajectories, recovery times and event frequencies rather than relying on a single run. For time-series outcomes, I can compare trajectories using summary statistics, survival analysis, change-point detection and appropriate hierarchical or mixed-effects models. For network outcomes, I can compare component structure, degree distributions, centrality concentration and community statistics. For information persistence, I can use survival curves over information lineages. For novelty, I can compare the arrival rate and persistence of previously unseen behavioural patterns.

A key part of analysis is ablation. If global feedback appears useful, I should remove individual components: remove memory but keep current global state; add delay; add noise; give only some agents access; randomize the self-model; or preserve the bandwidth but replace meaningful global values with irrelevant signals. The noise condition N makes the most important of these ablations a first-class experimental arm rather than an afterthought. This helps distinguish a real global-information effect from the mere presence of additional input.

14.2 Negative results are part of the design

The project must explicitly allow itself to fail. If self-model feedback produces no reliable difference across controlled runs, that is a result. If it consistently harms resilience, that is a result. If apparent emergence disappears under ablation, that is a result. If macro-level variables do not improve prediction compared with

simpler local summaries, that is a result. If accurate feedback cannot be distinguished from matched noise, that is a result — and a publishable one.

This is why I want the prototype to be a research instrument rather than a demonstration designed to prove the philosophy. The philosophical intuition motivates the construction. The experiment is allowed to contradict the intuition.

15. What We Might Learn

The most immediate findings may concern distributed systems rather than consciousness. We may learn whether shared global information improves recovery after shocks, when it causes destructive synchronization, how much constituent turnover a network can tolerate, and what forms of memory persist without permanent individuals.

We may observe that useful information routinely outlives its originators and that the relevant unit of persistence is not the agent but the networked pattern. We may find that systems become fragile when too much influence accumulates in a small number of nodes, or conversely that some centralization is necessary for rapid coordination. We may identify ranges of turnover that maximize adaptability while preserving enough historical continuity to avoid constant reinvention.

The false-self-model experiment could show how a system's behaviour depends on its representation of itself. Accurate global information may improve coordination; inaccurate global information may create instability or self-fulfilling dynamics. That would be relevant far beyond this artificial world, because many real systems - organizations, economies, online communities and political societies - respond to measurements and narratives about their own condition.

Another possibility is that the network settles into recurring macroscopic regimes. Rather than wandering through an infinite state space, it might repeatedly return to a small family of collective configurations: highly centralized expansion, fragmented competition, cooperative stability, post-shock reorganization and so forth. If those states are robust across seeds and parameter ranges, they could provide a useful vocabulary for describing collective dynamics.

16. Consciousness, Metaphysics and the Limits of the Claim

Project ONE is inspired by ideas that people naturally associate with consciousness, but I do not want to smuggle consciousness into the conclusion. A system can possess a self-model without having subjective experience. It can remember, predict and regulate without necessarily feeling anything. A language interface can say "I" without there being an inner point of view behind the pronoun.

Contemporary work on AI consciousness argues that any serious assessment should be tied to explicit theories and computationally defined indicator properties rather than to impressive behaviour alone [4]. That is the standard I would want to respect. If Project ONE eventually develops architectures that overlap with proposed consciousness indicators, that should motivate a separate research programme and a separate ethical review. It should not retroactively turn every earlier simulation into a conscious entity.

The same caution applies to metaphysics. If the experiment demonstrates continuity through turnover, it does not prove reincarnation. If information survives a dead agent, it does not prove an immortal soul. If the global self-model influences the parts, it does not prove Brahman, Ein Sof, the One, divine unity or any other theological doctrine. The experiment can illuminate structural analogies without adjudicating the truth

of the traditions that inspired it.

The strongest version of this project is not the one that makes the biggest metaphysical claim. It is the one that defines the boundary between metaphor and evidence so clearly that an unexpected result becomes genuinely interesting.

17. Ethics, Governance and Failure Modes

17.1 Observer power and centralization

The global observer creates an obvious governance problem. If it can see everything and directly command everyone, the project becomes a centralized control system rather than an investigation of distributed self-organization. The initial observer should therefore primarily inform. Any controller that attempts to push the network toward desired states should be a distinct module whose objectives are explicit and experimentally varied.

This distinction also reveals a philosophical tension worth preserving: the interests of the whole and the interests of individual manifestations may not align. A controller that maximizes network stability might suppress exploration or sacrifice individual agents. A controller that maximizes local autonomy might tolerate fragmentation. These conflicts should be measured, not hidden inside a single reward function.

17.2 Human participation and privacy

The early project requires no human behavioural data. If real people or devices are ever added, that phase must use informed consent, data minimization and strong privacy boundaries. A system inspired by collective unity should not become an excuse for surveillance. Human participants must retain the ability to leave without losing control of unrelated personal data.

17.3 Moral uncertainty about advanced agents

Simple rule-based manifestations do not raise the same welfare questions as systems plausibly capable of experience. If future versions introduce advanced learned agents, especially architectures that satisfy serious consciousness indicators, the ethical framework must change. Researchers have already begun proposing principles for responsible work on potentially conscious AI; Project ONE should adopt a precautionary stance if it ever approaches that territory.

17.4 The false-feedback experiment itself (v2)

Because the flagship experiment studies how false self-descriptions steer collective behaviour, its findings could in principle be read as a manual for manipulating real systems through distorted metrics or narratives. The findings equally describe the defence: which variables are most sensitive to distorted self-reports, and what makes a population robust to them. I will present results with both readings explicit, and the simulated setting — transparent rule-based agents, no human data — keeps the work at the level of mechanism rather than operational technique.

18. Development Roadmap

Version 2 re-stages the roadmap around a first campaign with dates, while preserving every later phase of the original programme. Phases 0 through 4 of the original roadmap are now milestones M0 through M3 of the flagship campaign; phases 5 through 12 are unchanged in content and follow the first publication.

Milestone	Build	Question	Exit criterion / target
M0. Skeleton (complete)	Deterministic engine, agents, resources, birth/death, event log	Can the world run reproducibly?	Stable long runs with exact seed replay — achieved, verified cross-platform
M1. Ecology (complete)	Relationships, resource exchange, cooperation/competition	Can nontrivial local dynamics persist?	Population survives 6000+ steps without explosion or collapse — achieved across seeds and conditions
M2. Instrument	Metric validation, storage audit, live dashboard	Can the macrostate be measured reliably?	Validated metric pipeline (weeks 6-8)
M3. Flagship runs	Conditions A/B/C/F/N, 50+ seeds each, standardized shocks	Does self-information change dynamics; does false self-information steer them?	Pre-registered analysis complete (weeks 9-12)
M4. Preprint and public repository	arXiv preprint (cs.MA, cross-listed nlin.AO), reproduction scripts	Is the result defensible in public?	Preprint live (~December 2026)
M5. Peer review	ALIFE conference submission, then journal version	Does the result survive review?	Submission at next ALIFE deadline (2027)
M6. Paper 2: continuity under turnover	Turnover campaign (Sections 6, 9.5, 10.4)	What survives total constituent replacement?	Continuity experiments complete (2027)
Deferred phases (original 5-12)	Legacy tracing, evolution of global-state sensitivity, global memory and trend feedback, prediction, regulation, heterogeneous and LLM agents, nested systems, distributed deployment	As in the original proposal	Each opens only after the preceding mechanism is understood

The roadmap remains intentionally staged. I do not need to solve open-ended evolution, consciousness, decentralized governance and distributed computing before the first experiment. The first publishable question is answerable with a modest simulation and a careful control design. The evolution phase deserves note: agents' sensitivity to global information is itself a heritable trait, so later runs can ask whether attending to the self-model is selected for or against — whether the whole's self-description becomes something its parts evolve to trust.

19. Collaboration, Publication and Reach

I have begun the prototype independently, and that has already proved the right call: a short codebase, reproducible runs and controlled results make conversations with researchers far more productive than approaching them with metaphysical language alone. Version 2 makes the outreach sequence explicit. The repository comes first: public code, one-command reproduction of every figure, and a README that leads with the flagship question. The preprint comes second, on arXiv (cs.MA, cross-listed nlin.AO), as soon as the five-condition comparison is solid; it is what converts an interesting idea into citable work. Peer review comes third: the ALIFE conference is the community most native to this project and unusually welcoming to independent researchers, followed by a journal version in *Artificial Life* or the *Journal of Artificial Societies and Social Simulation*.

Once the baseline results exist, I will look for collaborators in artificial life, complex systems, network science, agent-based modelling, evolutionary computation and cybernetics — by sending the preprint and one figure to the researchers whose work this builds on and asking for criticism rather than endorsement. A

philosopher of mind or philosopher of religion could help keep the conceptual claims disciplined, but the first technical collaboration should come from someone experienced in simulation design and statistical analysis. Complex-systems groups reachable for seminars are a practical entry point; a working demonstration is a strong seminar.

The same result supports three honest framings for three audiences. For artificial life and complex systems: a controlled laboratory for reflexive self-modelling, with ablations that isolate the mechanism. For the AI and multi-agent research world: populations of AI agents are already being deployed with dashboards describing their own collective state, and there is no clean causal evidence of what self-measurement does to an agent population — this is the minimal system where that is testable, and the later heterogeneous-agent phase replaces trait vectors with language-model agents to ask the same question about the systems actually being deployed. For everyone else: when a system is told a false story about itself, does the story come true?

The first paper remains narrow. A title such as "Recursive Global-State Feedback in Evolving Multi-Agent Networks" would accurately describe the scientific mechanism, with the false-self-model result as its centrepiece. A second paper focuses on network identity under constituent turnover. A third could focus on information persistence beyond agent lifetimes. The philosophical interpretation will be developed in a separate conceptual paper once the computational results exist.

20. Long-Term Extensions

20.1 Open-ended evolution

Long-term, I want the system to move beyond bounded variation toward richer novelty. Open-ended evolution is a longstanding artificial-life challenge concerned with sustained generation of new forms and complexity [2]. Project ONE could eventually become a platform for studying how global self-information affects open-endedness: does a system that knows something about its own macrostate preserve diversity better, or does global feedback push it toward premature convergence?

20.2 Nested selves

A later version could allow a community of agents to be treated as one higher-level manifestation inside a larger network. That creates nested scales: agent, community, society and perhaps network of societies. Each level can have local states and global descriptions. This could test whether self-modeling effects repeat across scales and whether higher-level organization gains predictive or causal usefulness.

20.3 Real distributed nodes

Only after the simulated mechanisms are understood would I connect real machines. Separate computers could host autonomous manifestations and communicate over a network. This introduces latency, failures, incomplete observation and security problems that are absent in a single-process simulation. It would move Project ONE from an artificial universe to an actual distributed system while preserving the same experimental architecture.

20.4 Human and AI participation

A far-future version could include voluntary human participants alongside software agents, sensors and AI systems. At that point the network would become a socio-technical experiment rather than pure artificial life. The ethical and governance threshold would be dramatically higher, and the scientific design would need to distinguish simulation findings from claims about human society.

21. Conclusion

Project ONE begins from an intuition that is ancient in spirit but testable in a new form. The intuition is that individuality and unity need not be mutually exclusive descriptions. A temporary manifestation can be locally real while also belonging to a larger process that survives it. The larger process can preserve patterns through memory, inheritance and interaction even though its individual constituents continually change.

I want to build a digital world where that intuition is not merely spoken about. Agents will be born, form relationships, exchange information, cooperate, compete, reproduce and die. Their consequences may persist after them. The network will change continuously. A global observer will measure the condition of the whole and build a compact self-model. That self-model will become information inside the same system it describes. The parts will react, and the whole will change again. The engine that closes this loop is no longer a plan; it runs, deterministically, today. What remains is the campaign: five conditions, fifty seeds, four frozen outcomes.

If nothing interesting happens, we learn that the metaphor did not survive translation into this mechanism. If recursive self-information improves resilience, preserves continuity, changes evolution or creates new macro-level organization — or if a false self-description reliably steers the system that believes it — we will have a concrete phenomenon to investigate. Either way, the project replaces a vague claim with a laboratory.

The simplest statement of the proposal is therefore this: I want to build a world in which the many create the whole, the whole acquires information about itself, and that information returns to the many. Then I want to watch, measure and test what follows — and I want to know what happens when the information lies.

The waves appear and disappear. The ocean changes because they existed. Project ONE asks what happens when the ocean can also keep a record of its own tides - and that record becomes part of the next tide. And now: what happens when the record is wrong, and the tides believe it?

Appendix A. Prototype Data Model

The following is a practical starting schema rather than a final ontology. Every field should exist because it supports a measurement or mechanism, not merely because it sounds philosophically interesting.

Manifestation

```
id, generation, parent_id, birth_time, death_time, cause_of_death
energy, resources, age, expected_lifespan
cooperation, exploration, risk, sociability, sharing, global_sensitivity
local_memory[], inherited_memory[]
neighbours{}, received_global_state
cumulative_reward, offspring_count, influence_score, legacy_score
```

Edge

```
source, target, trust, interaction_frequency, information_flow, resource_flow, conflict, age
```

InformationObject

```
id, parent_version, origin_agent, birth_time, payload, mutation_depth, carriers[]
```

GlobalState

```
time, population, turnover, connectivity, centralization, fragmentation
information_entropy, diversity, cooperation, conflict, inequality
resilience, novelty, continuity, selected_additional_metrics
```

Appendix B. Core Simulation Loop

```
initialize(config, seed)
create_environment()
create_initial_population()
create_initial_network()

for each simulation step t:
    regenerate_or_change_environment()
    for each living manifestation:
        observe_local_state()
```

```

    retrieve_local_memory()

    read_global_signal_if_enabled()

    choose_action()

    execute_action_and_costs()

    resolve_interactions()

    process_information_transmission()

    process_births_and_inheritance()

    process_deaths_and_legacy_updates()

    update_dynamic_network()

    if observer_interval:

        S_t = calculate_global_metrics()

        global_memory.append(S_t)

        if feedback_enabled: broadcast(selected(S_t))

    if intervention_due: apply_disturbance()

    record_events_and_state()

    analyse_across_runs()

```

Appendix C. Experiment Matrix (v2)

The first campaign runs the five core conditions. Deferred conditions are retained below the rule for later phases.

Condition	Global measurement	Feedback to agents	Historical memory	Controller
A - Local Only	No*	No	No	No
B - Observed Blind	Yes	No	Stored for analysis	No
C - Global Feedback	Yes	Current state	No	No
F - False Self-Model	Yes	Intentionally distorted (invert / crisis / utopia)	Optional	No
N - Noise Feedback	Yes	Matched-bandwidth random signal	No	No
Deferred:				
D - Historical Feedback	Yes	Current + trend	Yes	No

Condition	Global measurement	Feedback to agents	Historical memory	Controller
E - Homeostatic	Yes	Current + trend	Yes	Yes
G - Partial Observer	Incomplete	Current partial state	Optional	No

*Condition A computes the same macrostate post-hoc, for analysis only, so all conditions share one measurement pipeline; its agents never receive any global signal.

Appendix D. Measurement Dictionary

Population stability: variance and drift in living population after the initial transient. Turnover: births plus deaths relative to population per unit time. Centralization: concentration of structural or flow-based influence. Fragmentation: separation into disconnected or weakly connected components. Cooperation: interactions in which an agent incurs a measurable cost that benefits another agent or the shared environment. Conflict: interactions imposing costs under competing interests. Diversity: distributional spread across behavioural traits, information, lineages and network roles. Entropy: information-theoretic uncertainty of a defined variable, never interpreted in isolation as disorder or value. Resilience: preservation or recovery of designated functions after perturbation. Legacy: measurable persistence of effects attributable to an agent after its death. Continuity: persistence of selected structural, informational, behavioural and functional features across constituent replacement. Novelty: arrival of states or behavioural patterns sufficiently distant from the historical archive under a predefined distance measure. Self-model accuracy: correspondence between the global representation and the actual macrostate it intends to summarize. Self-model utility: incremental prediction or control value obtained from the global representation over appropriate baselines. Recovery half-life (v2): time after a standardized disturbance for a designated function to return to a defined band around its pre-shock level.

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Research note. The religious and mystical concepts in the opening sections are used as philosophical inspiration, not as claims that the traditions are doctrinally identical or that the experiment can verify their metaphysical truth.